Bonds and Stocks:
Twenty Years of Relative Market Size Across Ten Economies

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Abstract

We assemble a panel of bond market outstanding and equity market capitalization for ten economies from 2006 to 2025 and track the bond-to-equity ratio, the growth of each market, and their relative volatility. Bond outstanding exceeds equity market capitalization in nine of the ten economies. The United States is the sole large economy where equities overtook bonds: the ratio fell from 1.22 in 2006 to 0.89 in 2025. Everywhere else bonds dominate, and the gap widened or held: Italy (4.12), Brazil (3.93), Germany (2.00), France (2.13 in 2018), China (1.85), Japan (1.68). China's bond market grew twenty-four-fold, from $1.2 trillion to $28.7 trillion. Equities are two to seven times more volatile than bonds in every country; the ratio itself moves almost entirely on the equity leg. In 2008, equities fell 28 to 60 percent while bond outstanding grew in most economies, spiking every ratio. The government share of bond outstanding rose in every country studied. Aggregating to the world level, global bond outstanding (BIS, 42 reporting countries, nominal) reached $140.6 trillion in 2024 against $115.1 trillion of listed equity market capitalization (World Bank), a ratio of 1.22; bonds grew 5.8 percent per year from 2006 to 2024 against 4.7 percent for equities. The global ratio never falls to the US level because the US, 54 percent of world equities, is the outlier: excluding America, the world ratio is 1.56. A central measurement caveat: bond series mix market value and nominal valuations across countries, which we document explicitly and which limits the precision of level comparisons.

Keywords: bond market, equity market capitalization, financial structure, market volatility, government debt.
JEL codes: G10, G15, E44, F30.

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1 Introduction

The bond market is larger than most people realize. In the United States it is roughly the same size as the equity market, and in most other large economies it is substantially larger, yet it receives a fraction of the public and media attention that equities command. This paper asks a simple question across ten economies over twenty years: how big is the bond market relative to the equity market, how has that relationship changed, and which of the two is more volatile?

We build the comparison from primary sources. Bond market outstanding comes from the Bank for International Settlements debt securities statistics and from national sources (central banks, statistical offices, treasuries) where BIS coverage is incomplete. Equity market capitalization comes from the World Bank World Development Indicators, cross-checked against the World Federation of Exchanges and national exchange data. The panel covers the United States, China, Japan, Germany, India, the United Kingdom, France, Italy, Canada, and Brazil from 2006 through 2025, 180 country-year observations.

Three findings stand out. First, the United States is the exception, not the rule. It is the only large economy in the panel where equities overtook bonds: the bond-to-equity ratio fell from 1.22 in 2006 to 0.89 in 2025, driven by the post-2020 equity bull market. In the other nine economies, bonds exceed equities, and the gap widened or held steady in most of them. Italy's ratio reached 4.12 and Brazil's 3.93 by 2024.

Second, China's bond market is the great buildout of the period. It grew from $1.2 trillion to $28.7 trillion, a twenty-four-fold increase, while its equity market grew thirteen-fold. The ratio rose from 1.05 to 1.85.

Third, the two markets differ sharply in volatility. The standard deviation of annual equity market growth is two to seven times that of bond market growth in every country (18.0 percent versus 2.9 percent in the United States; 72.6 percent versus 10.0 percent in China). The bond-to-equity ratio therefore moves almost entirely on the equity leg. In 2008, equities fell 28 to 60 percent across the panel while bond outstanding grew in most economies, and every country's ratio spiked.

The rest of the paper proceeds as follows. Section 2 describes the institutional measurement of bond and equity markets. Section 3 reviews related literature. Section 4 documents the data, including a central valuation caveat that constrains the precision of cross-country level comparisons. Section 5 describes the computations. Section 6 presents results: the ratio over time for all ten economies, the volatility analysis, the 2008 and 2020 episodes, the government-share shift, and cross-country patterns. Section 7 states the limitations. Section 8 concludes. References and a data appendix follow.

2 Institutional background

A bond market's outstanding stock is the face or market value of all debt securities issued by residents and not yet matured: government bonds, corporate bonds, and in some measures asset-backed securities. It grows through new issuance and shrinks through maturities and defaults. An equity market's capitalization is the market value of all listed domestic companies' shares. It grows through price appreciation and new listings and shrinks through price declines and delistings.

The two measures differ in what moves them. Bond outstanding is driven primarily by issuance decisions: governments borrowing to fund deficits and corporations borrowing to fund investment. It rarely falls sharply, because maturing bonds are normally refinanced. Equity capitalization is driven primarily by prices, which revalue the entire stock of listed shares every trading day. This structural difference is the reason for the volatility gap documented in Section 6.

Valuation matters for the comparison. Equity capitalization is always a market value. Bond

outstanding is reported at nominal (face) value in some sources and at market value in others. When interest rates rise, the market value of outstanding bonds falls below nominal value; when rates fall, it rises above. Mixing the two across countries introduces measurement error into level comparisons, which we document in Section 4 and carry through the interpretation.

3 Related literature

The comparative size of bond and equity markets has a long literature in financial structure. Levine (2002) surveys the bank-based versus market-based financial systems debate, in which bond markets occupy an intermediate position. Demirguc-Kunt and Levine (2001), in the World Bank's financial structure database work, document cross-country variation in the relative size of banking, bond, and equity markets. The BIS publishes the most comprehensive cross-country debt securities statistics (BIS, 2024), and the World Bank's World Development Indicators remain the standard source for cross-country equity market capitalization.

On the growth of bond markets specifically, Turner (2014) documents the post-crisis expansion of bond financing as bank lending retreated, and Greenwood, Hanson, and Stein (2010) analyze the gap-filling behavior of corporate issuers. For China, the rise of the domestic bond market is documented in Amstad, Sun, and Xiong (2019). On volatility, the equity premium and excess volatility literature (Shiller, 1981) establishes that equity prices move far more than fundamentals, which is consistent with the volatility gap we measure between the two markets' aggregate size.

4 Data

4.1 Sources and coverage

Table 1 lists the bond data source for each country. For the United States, China, and Canada we use the BIS debt securities statistics, total economy, debt securities, all maturities, liabilities, in US dollars at nominal value, which validates against published totals (US 2024: $58.2 trillion, matching SIFMA). For Germany, France, and Italy we use Eurostat financial balance sheets and national central bank data at market value. For Japan we use the Bank of Japan Flow of Funds at face value. For the United Kingdom we use the Office for National Statistics UK Economic Accounts. For Brazil we use the Tesouro Nacional for government debt and BIS for the 2021+ total. For India we use the Reserve Bank of India Handbook of Statistics and SEBI for corporate bonds.

Equity market capitalization is the World Bank World Development Indicator CM.MKT.LCAP.CD (domestic market capitalization, USD), retrieved 2026-10-07, last updated 2026-07-13 per the API. Gaps for France (2019-2025), the United Kingdom (2020, 2023-2025), and Italy (2015-2018) were filled where possible from Borsa Italiana, ECB, and WFE data; remaining gaps are documented in the panel and the affected ratios are not computed for missing years. The US 2024 figure ($62.19 trillion) matches the SIFMA Fact Book.

4.2 The valuation caveat

This is the central limitation of the exercise and we state it plainly. Bond outstanding is measured at *market value* for Germany, Italy, France, the United Kingdom, and the Brazil total, and at *nominal or face value* for the United States, China, Canada, Japan, India, and Brazilian government debt. Equity market capitalization is always a market value. The mixture is forced by source availability: no single valuation is available for all ten countries over the full period.

The consequence is that level comparisons across the two valuation groups carry measurement error whose sign depends on the interest rate environment. In the high-rate environment of

Table 1: Bond outstanding sources by country

Country	Source	Valuation	Vintage
United States	BIS debt securities statistics	Nominal	Sep 2026
China	BIS debt securities statistics	Nominal	Sep 2026
Japan	Bank of Japan, Flow of Funds	Face value	2026-09-17
Germany	Eurostat financial balance sheets	Market value	2026-10-07
India	RBI Handbook of Statistics; SEBI	Face value	2026-07-31
United Kingdom	ONS UK Economic Accounts; BIS	Market value	2026-09-30
France	Eurostat financial balance sheets	Market value	2026-10-07
Italy	Banca d'Italia	Market value	2026-07-14
Canada	BIS debt securities statistics	Nominal	Sep 2026
Brazil	Tesouro Nacional; BIS	Mixed	2026-10-08

Table 2: Summary statistics, bond-to-equity ratio, 2006-latest

Country	First year	Last year	Ratio, first	Ratio, last
United States	2006	2025	1.22	0.89
China	2006	2025	1.05	1.85
Japan	2006	2024	1.96	1.68
Germany	2006	2024	2.39	2.00
India	2006	2024	0.34	0.54
United Kingdom	2006	2022	1.03	1.57
France	2006	2018	1.26	2.13
Italy	2006	2024	2.72	4.12
Canada	2006	2025	0.78	1.01
Brazil	2006	2024	1.02	3.93

2023-2025, market values sit below nominal values, so market-value bond series understate outstanding relative to nominal-value series, biasing those countries' ratios downward. The bias reverses in low-rate environments. Within a country, the valuation is consistent over time, so each country's own trajectory is unaffected. We flag the valuation per observation in the panel dataset and advise readers to weight trend comparisons more heavily than level comparisons across valuation groups.

Two further estimates are flagged in the data. Brazil's 2006-2020 bond total is estimated by interpolating the government share from 0.80 in 2006 to the observed 0.504 in 2021; India's 2006-2009 total is government debt only, as no corporate bond series exists before 2010, and is therefore understated. Both are marked with an estimation flag in the panel.

All figures are in US dollars at year-end exchange rates. We do not decompose currency effects: a weaker euro mechanically shrinks Germany's dollar-denominated bond stock, for example. The ratios are therefore affected by FX moves, which we note but do not adjust.

5 Methodology

For each country c and year t we compute the bond-to-equity ratio

$$R_{c,t} = \frac{B_{c,t}}{E_{c,t}}, \tag{1}$$

where B is bond outstanding and E is equity market capitalization, both in US dollars. Annual growth rates are $g^B_{c,t} = B_{c,t}/B_{c,t-1} - 1$ and analogously for equities. Volatility is the sample standard deviation of annual growth rates over the available period. The government share is government debt securities outstanding divided by total bond outstanding. All computations use the panel in `data/panel_10c.csv`; every figure in Section 6 is produced directly from it.

6 Results

6.1 Market size in 2024

Figure 1 shows the absolute scale. The United States dominates both markets: $58.2 trillion in bonds and $62.2 trillion in equities in 2024. China is second in bonds at $25.0 trillion, nearly half the US level, and second in equities at $11.7 trillion. Japan's bond market ($10.6 trillion) is the third largest, reflecting decades of government issuance. India is the most equity-tilted large economy in absolute terms: $5.1 trillion in equities against $2.8 trillion in bonds.

6.2 The ratio over time

Figure 2 plots the bond-to-equity ratio for all ten economies. We discuss each in turn, in alphabetical order.

Brazil. The ratio rose from 1.02 in 2006 to 3.93 in 2024, the sharpest increase in the panel. Bonds grew from $0.7 trillion to $2.6 trillion (3.6-fold) while equities shrank 7 percent, from $0.7 trillion to $0.7 trillion in rounded terms. Brazil is the only country where the equity market ended the period smaller in dollar terms than it began it. The government share of bonds fell from 80 to 46 percent, the only decline in the panel, as private issuance grew faster than public. The 2006-2020 bond total is partly estimated; see Section 4.

Canada. The ratio rose from 0.78 to 1.01, crossing parity around 2015. Both markets roughly tripled: bonds $1.3 trillion to $4.6 trillion, equities $1.7 trillion to $4.6 trillion. The government share fell from 54 to 45 percent.

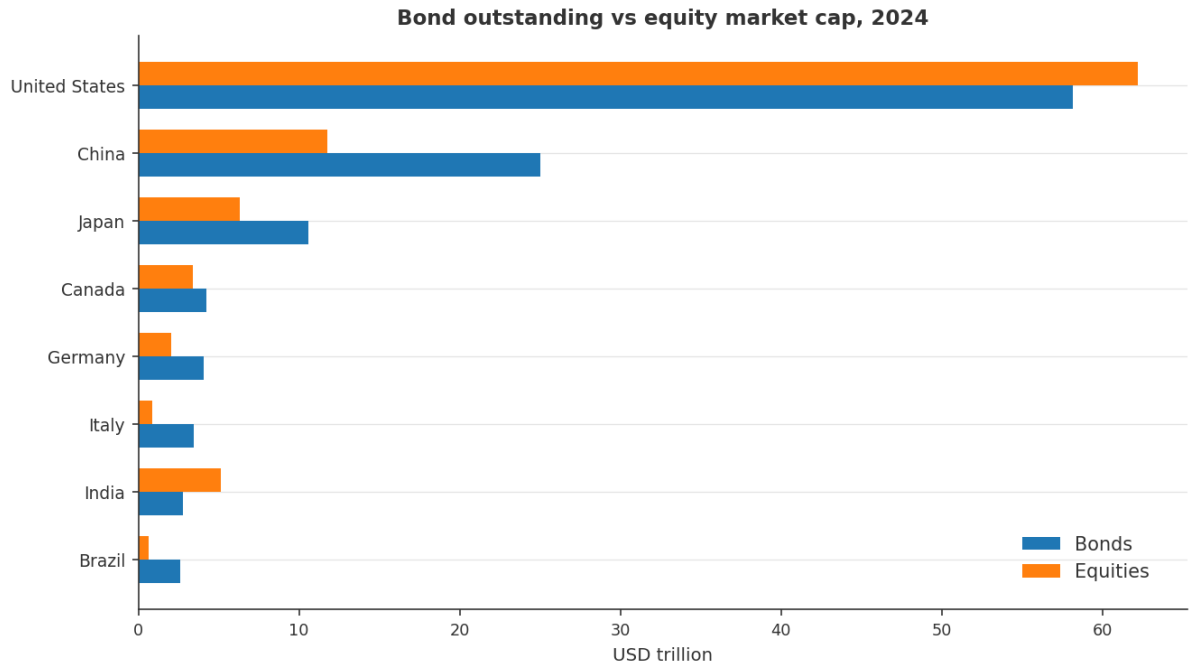


Figure 1: Bond outstanding versus equity market capitalization, 2024, USD trillion. The United States is the largest in both markets; China is second in both. India is the only large economy where equities clearly exceed bonds in absolute terms. *Source:* BIS debt securities statistics; Eurostat; Bank of Japan; ONS; Banca d'Italia; Tesouro Nacional; RBI; SEBI; World Bank WDI. See Table 1 for vintages. Valuation mixes market and nominal value across countries; see Section 4.

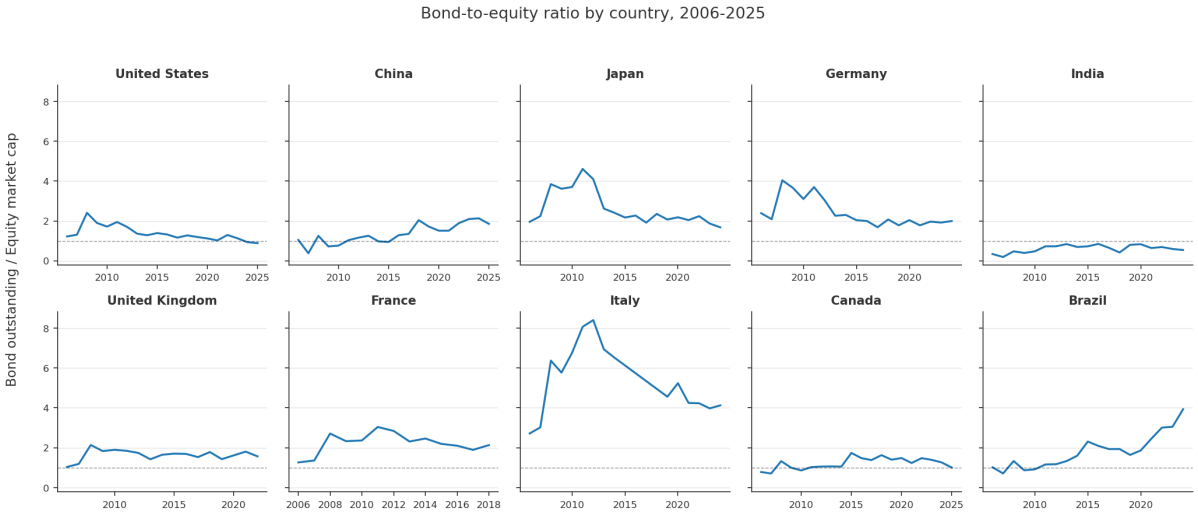


Figure 2: Bond-to-equity ratio by country, 2006-2025. Dashed line marks parity. The United States is the only economy where the ratio fell below one; Italy and Brazil show the largest bond dominance. *Source:* as in Figure 1. France ends 2018 and the United Kingdom 2022 on equity data availability.

China. The ratio rose from 1.05 to 1.85. Bonds grew twenty-four-fold, $1.2 trillion to $28.7 trillion, the fastest bond growth in the panel; equities grew thirteen-fold, $1.1 trillion to $15.5 trillion. The government share rose from 32 to 49 percent. The 2007 equity spike, when the ratio collapsed to 0.38, and the subsequent recovery illustrate how equity volatility dominates the ratio's path.

Germany. The ratio fell from 2.39 to 2.00, the highest starting level among major economies and still the most bond-dominated of the large European economies at the end. Bonds were essentially flat in dollar terms ($3.9 trillion to $4.1 trillion); equities grew modestly ($1.6 trillion to $2.0 trillion). The government share rose from 38 to 51 percent.

India. The ratio rose from 0.34 to 0.54, the lowest in the panel throughout: India is the most equity-tilted economy studied. Bonds grew nine-fold ($0.3 trillion to $2.8 trillion), equities nearly six-fold ($0.9 trillion to $5.1 trillion). The early bond figures are government-only and therefore understated; the true 2006 ratio was higher than 0.34, which strengthens rather than weakens the finding that bonds are catching up.

Italy. The ratio rose from 2.72 to 4.12, the highest ending level in the panel. The 2008-2012 spike to 8.4 reflects the sovereign debt crisis: equities collapsed while government issuance surged. Bonds grew modestly ($2.8 trillion to $3.5 trillion); equities shrank ($1.0 trillion to $0.8 trillion). The government share, already high at 63 percent, rose to 77 percent.

Japan. The ratio fell from 1.96 to 1.68. Japan remains structurally bond-heavy: $10.6 trillion in bonds against $6.3 trillion in equities in 2024, with the government share rising from 64 to 75 percent, the highest in the panel. Both markets grew slowly; the ratio's decline reflects steady equity appreciation against a bond stock that grew only 1.2-fold.

United Kingdom. The ratio rose from 1.03 to 1.57 (2022, latest available). Bonds grew from $3.9 trillion to $4.8 trillion; equities shrank from $3.8 trillion to $3.1 trillion. The government share doubled from 25 to 50 percent, the largest increase in the panel, reflecting post-crisis and pandemic issuance.

France. The ratio rose from 1.26 in 2006 to 2.13 in 2018, the last year with equity data. Bonds grew 1.6-fold; equities were flat. The government share rose from 46 to 52 percent. The truncated series is the panel's main coverage gap.

United States. The ratio fell from 1.22 to 0.89, the only decline to below parity in the panel. Bonds grew 2.6-fold ($23.9 trillion to $61.1 trillion); equities grew 3.5-fold ($19.6 trillion to $68.9 trillion). Bonds exceeded equities until about 2020; the post-2020 equity bull market flipped the ratio. The government share rose from 30 to 55 percent, meaning the majority of the growth in US bond outstanding was public borrowing.

6.3 Volatility

Figure 3 compares the standard deviation of annual growth for the two markets. Equities are more volatile than bonds in every country, by factors ranging from 1.4 (Japan) to 7.3 (China). The United States shows the calmest bond market in the panel (2.9 percent) against equity volatility of 18.0 percent. China's equity volatility of 72.6 percent reflects the 2007 boom-bust cycle.

Because bond outstanding moves slowly, the bond-to-equity ratio moves almost entirely on the equity leg. Regressing annual changes in the ratio on the two growth rates (pooled,

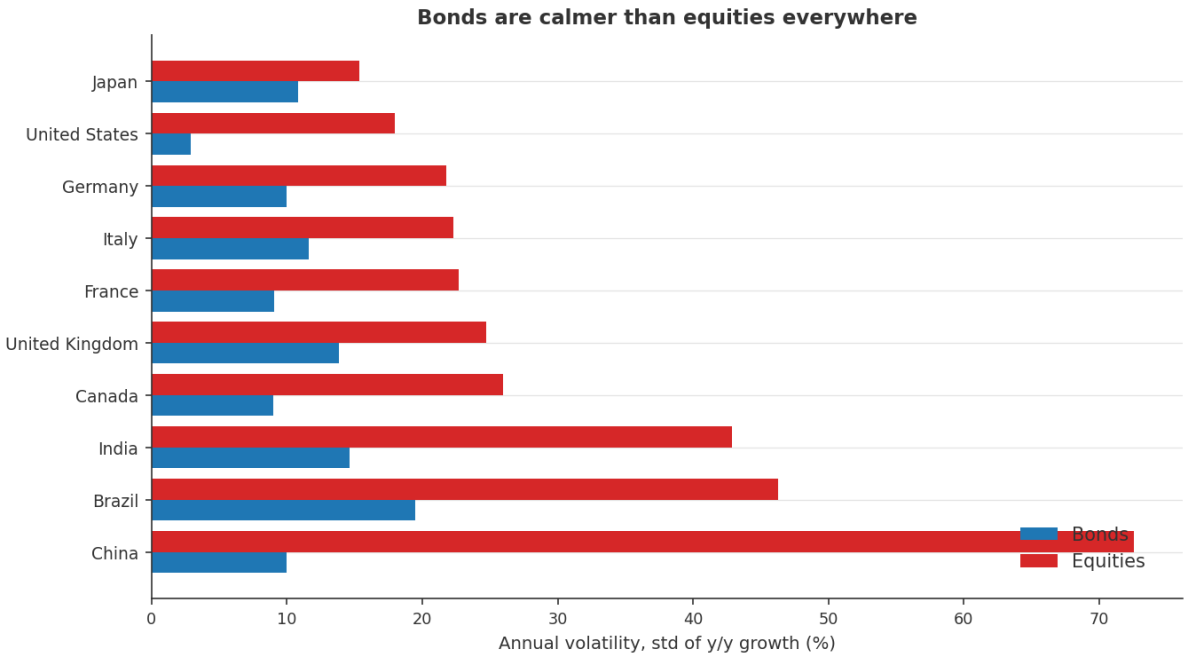


Figure 3: Annual volatility, standard deviation of year-on-year growth, 2006-latest. Equities are more volatile than bonds in all ten economies. *Source:* panel dataset; see Section 5.

170 observations) gives a coefficient of -0.94 on equity growth and 0.31 on bond growth: a 10 percent equity rally cuts the ratio by roughly 9 percent, while a 10 percent expansion in bond outstanding raises it by roughly 3 percent.

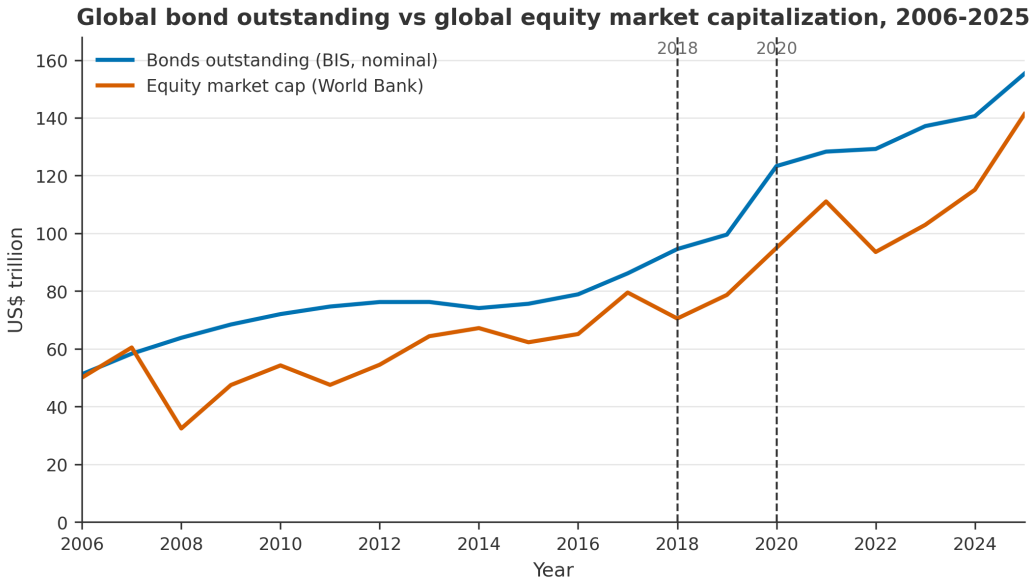
6.4 Crisis episodes

2008. Equities fell 28 to 60 percent across the panel in 2008. Bond outstanding grew in most economies (the exceptions were Brazil, Canada, and the United Kingdom, where dollar-denominated figures fell on currency moves). Every country's ratio spiked: the US from 1.31 to 2.41, Germany from 2.09 to 4.04, Italy from 3.02 to 6.37, Japan from 2.24 to 3.85. The mechanism is visible in the data: flight to safety supported bond valuations and issuance continued, while equity prices collapsed. The ratio is therefore countercyclical, rising hardest exactly when equities fall.

2020. Both markets grew in 2020 on fiscal and monetary stimulus: US bonds up 2.4 percent and equities up 30 percent is illustrative. Ratios generally fell or held steady, the mirror image of 2008. The two crises together show that the ratio's crisis behavior depends on whether the shock hits bond issuance and equity prices in the same direction.

6.5 The government share

The government share of bond outstanding rose in nine of the ten countries (Brazil excepted). The increases are large: the US from 30 to 55 percent, the UK from 25 to 50 percent, Germany from 38 to 51 percent, Japan from 64 to 75 percent, Italy from 63 to 77 percent. The post-2008 and post-2020 fiscal expansions are the cause. In other words, much of the growth in global bond outstanding over the last twenty years is government borrowing, not corporate.



Dashed lines mark BIS reporting-coverage expansions: 2018 (+5 countries incl. Brazil, Korea, Mexico) and 2020 (+7 countries incl. the United Kingdom).

Figure 4: Global bond outstanding (BIS debt securities statistics, nominal, 42 reporting countries) and global listed equity market capitalization (World Bank WDI). Dashed lines mark BIS reporting-coverage expansions in 2018 (+5 countries) and 2020 (+7 countries); see text.

6.6 Cross-country patterns

Three patterns emerge. First, bank-based financial systems (Germany, Italy, Japan) are bond-heavy; market-based systems (the US, and to a lesser extent India and Canada) are equity-tilted. Second, the direction of travel is not uniform: the US ratio fell while most others rose, so cross-country dispersion increased. Third, emerging markets diverge: China's ratio rose on the great bond buildout, India's stayed the lowest in the panel, and Brazil's rose because its equity market shrank in dollar terms.

6.7 The global picture

The country results aggregate cleanly. Summing BIS debt securities statistics across all 42 reporting countries gives global bond outstanding of $51.2 trillion in 2006, $140.6 trillion in 2024, and $155.3 trillion in 2025 (nominal, US dollars; Figure 4). The 2023 figure, $137.2 trillion, sits just below ICMA's independent $140 trillion estimate for that year, which suggests the BIS reporting population captures nearly the whole market; the residual gap is countries outside BIS reporting, principally India, Australia, Russia, and Saudi Arabia. World Bank data put global listed equity market capitalization at $50.1 trillion in 2006, $115.1 trillion in 2024, and $141.3 trillion in 2025.

Three facts stand out. First, the global bond-to-equity ratio has been above one for almost the entire period: 1.02 in 2006, spiking to 1.97 in 2008 when global equities lost 46 percent of their value while bond outstanding kept growing, then settling between 1.08 and 1.38 from 2013 onward (1.22 in 2024, 1.10 in 2025; Figure 5). Bonds are the larger asset class at the world level, not just in most individual countries. Second, bonds grew faster than equities over the full period: 5.8 percent per year versus 4.7 percent from 2006 to 2024 in nominal dollar terms. The 2020 COVID borrowing surge is visible in both the level and the ratio. Third, the global ratio is much lower than the typical country's because the United States dominates global equities: the US accounts for 54 percent of world market capitalization but 41 percent of world bond outstanding (2024). Excluding the US, the global ratio rises from 0.90 in 2006 to 1.56 in 2024.

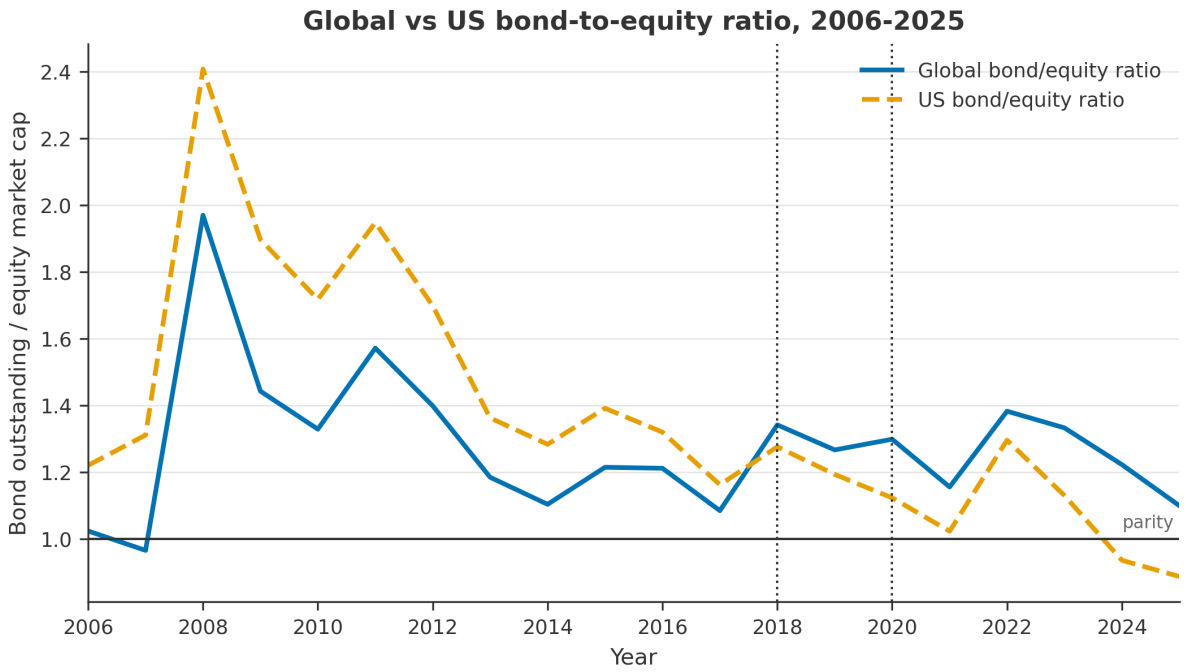


Figure 5: Global bond-to-equity ratio against the US ratio. The US crosses below parity after 2020; the global ratio never does. Dotted lines mark the 2018 and 2020 BIS coverage expansions.

The world outside America is unambiguously bond-heavy, and becoming more so; the US is the outlier that drags the global average toward parity.

Two measurement notes qualify the global series. The BIS reporting population expanded in 2018 (adding Brazil, Korea, Mexico, Indonesia, and Argentina, $5.8 trillion, 6.1 percent of the total) and 2020 (adding the United Kingdom, Sweden, Poland, Denmark, and others, $8.0 trillion, 6.4 percent), so part of the level growth in those years is coverage, not issuance; the ratio moves less than the level because the added countries carry both bonds and, in the equity series, were already counted. And the global bond aggregate is entirely nominal (BIS VAL=N, the one valuation with consistent cross-country coverage), while equities are market value: the same valuation caveat as the country panel, applied uniformly rather than mixed.

7 Limitations and robustness

The valuation caveat in Section 4 is the main limitation: level comparisons across the market-value and nominal-value groups carry error that varies with the rate cycle. The within-country trends, the volatility ranking, and the crisis-episode findings do not depend on it.

Three further limits apply. First, the Brazil 2006-2020 and India 2006-2009 bond figures are estimated or partial, as documented; they affect early-period levels for those two countries but not the direction of their trends. Second, France's equity series ends in 2018 and the United Kingdom's has gaps, so their recent ratios are not observed. Third, all figures are in US dollars without FX decomposition, so currency moves contaminate the growth rates; the 2008 declines in Brazil, Canada, and the UK bond series are partly currency effects.

A natural falsifier: if bond markets were merely keeping pace with nominal GDP while equities captured real growth, the ratio's decline in the US could be a denominator story about equity valuations rather than a numerator story about bond issuance. The data reject the strong form of this: US bond outstanding grew 2.6-fold in nominal terms over the period, far

above cumulative nominal GDP growth, and the government share rose from 30 to 55 percent. Issuance, not just valuation, drove the numerator.

8 Conclusion

Across ten economies and twenty years, the bond market is bigger than the equity market almost everywhere, calmer than the equity market everywhere, and increasingly dominated by government borrowing. The United States is the exception that proves the pattern: the only large economy where equities overtook bonds, and only after the post-2020 bull market. China's twenty-four-fold bond expansion is the period's largest structural change. The bond-to-equity ratio is countercyclical, spiking when equities crash, because bond outstanding rarely falls. For anyone allocating across asset classes or assessing financial structure, the bond market's quiet scale is the fact the data keep returning. At the world level the verdict is the same as in nine of the ten countries: bonds are bigger, calmer, and growing faster than equities, and the one economy that breaks the pattern is the United States.

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Data appendix

The panel dataset is `data/panel_10c.csv` (180 rows: 10 countries, 2006-2025 where available). Columns:

- `country`: ISO3 code.
- `year`: calendar year (India: fiscal year starting April, mapped to starting calendar year).
- `bond_total_usd_bn`: total debt securities outstanding, liabilities of residents, USD billion, year-end.
- `bond_govt_usd_bn`: government component, USD billion.
- `source`: issuing source (BIS or national).
- `est_flag`: TRUE for Brazil 2006-2020 totals (interpolated government share) and India 2006-2009 totals (government only).
- `mktcap_usd_bn`: World Bank WDI market capitalization of listed domestic companies, current USD, billion.
- `ratio`: `bond_total_usd_bn` / `mktcap_usd_bn`.
- `govt_share`: `bond_govt_usd_bn` / `bond_total_usd_bn`.
- `bond_growth`, `eq_growth`: annual growth rates.

Valuation per country is listed in Table 1. FX conversion uses ECB euro reference rates at year-end (December, or March for India). Producer-first citations: BIS, World Bank, Eurostat, Bank of Japan, ONS, Banca d'Italia, Tesouro Nacional, RBI, SEBI, WFE.

The global aggregates are `data/global_aggregates.csv` (20 rows, 2006-2025). `global_bond_usd_tn` sums BIS DSS total debt securities (liabilities, all issuers S1, nominal VAL=N, USD, Q4) across 42 reporting countries, excluding the U2 euro-area aggregate to avoid double counting; `bis_reporting_countries` records the reporting population each year. `global_equity_usd_tn` is World Bank WDI series CM.MKT.LCAP.CD for the WLD aggregate, and `global_ratio` is their quotient.

Production note: This paper was produced with AI assistance. All quantitative claims trace to the panel dataset, the global aggregates dataset, or to cited primary sources listed above.